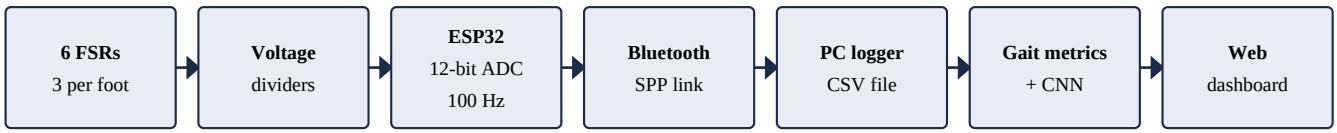
Smart Gait Recognition Insole

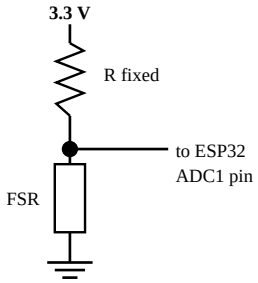
Embedded Systems Course Project | Project brief for the review panel

1. Objective

A low-cost wearable that measures foot pressure with six force-sensitive resistors (FSRs) and an ESP32, analyses gait from the signals, and presents the findings in a clinic-style web dashboard. Three tasks: **(1)** AI gait recognition, **(2)** evaluation of gait features linked to Parkinson's disease and stroke (step rate, variability, left-right asymmetry), and **(3)** a smart healthcare IoT dashboard.



2. Hardware and circuit



Sensing circuit. Each FSR's resistance falls as load rises. It forms a voltage divider with a fixed resistor: the divider node is read by the ESP32's 12-bit ADC (0 to 4095). Unloaded the node sits near 4095 (FSR very high resistance); pressing pulls it down, and a hard press can saturate the pad near 0.

All six pads use **ADC1** pins only, because ADC2 is shared with the Wi-Fi radio and the input-only pins 34, 35, 36 and 39 suit analog sensing. The pin map below was verified by pressing each pad.

Channel	L toe	L mid	L heel	R toe	R mid	R heel
ESP32 GPIO	32	33	34	39	36	35

3. Firmware and data acquisition

Arduino firmware on a DOIT ESP32 DevKit V1: a FreeRTOS task pinned to core 1 samples all six channels every 10 ms (**100 Hz**, fixed period with *vTaskDelayUntil*) and pushes them into a 500-sample queue. The main loop batches rows and sends them as CSV text (*n, t_ms, six channels*) over **Bluetooth Classic SPP** (device name "insole"); the queue decouples sampling from radio delays, so a slow Bluetooth moment does not disturb the sampling clock. A Python logger on the PC saves each session as a CSV and prints a quality report (rate, gaps, saturated or dead pads).

4. Signal processing and gait features

- **Calibration:** a standard load gives per-pad conductance, G = (4095 – x) / x, used for the pressure map and load tables. Pads reading at or below 150 counts are flagged saturated.
- **Walking:** stance and swing are found per foot with hysteresis thresholds; stride time, stride variability (CV) and stance asymmetry follow.
- **Stepping in place:** the difference of the two foot signals (each scaled to its own range) changes sign as the feet take turns; a sign change is a step. Results are checked at three detection thresholds and labelled stable or uncertain.

5. Machine learning (Task 1)

Stage	Detail
Data	PhysioNet gait-in-Parkinson's dataset (gaitpdb): force-sensor walking records, Parkinson's versus healthy. 3 of its 8 sensors per foot (heel, mid, toe) are matched to our 3 pads. 28,069 two-second windows.
Preprocessing	Load signal = drop below the unloaded ADC level; channels reordered to the training order; trim 2 s at each end of long walks; z-score each channel per recording; 100 Hz to 50 Hz; 2 s windows (6 channels x 100 samples), 50% overlap.
Model	1-D CNN: three Conv1D blocks (32, 64, 128 filters; kernels 7, 5, 3), each with batch norm, ReLU and max-pool; global average pool; linear layer to 2 classes. 37,090 parameters.
Training	15 epochs, AdamW (lr 1e-3, weight decay 1e-4), cosine schedule, class-weighted cross-entropy. Validation: subject-wise 5-fold macro F1 0.759; leave-one-study-out 0.706.
Inference	Weights exported to NumPy, so the dashboard server needs no PyTorch. Output = mean Parkinson's-like probability over all windows of a recording.

6. Web dashboard (Task 3)

FastAPI back end with SQLite storage; single-page front end in plain JavaScript and SVG. Features: patient records, bulk CSV import, pressure map with playback, full-length trace, sensor-quality card, trends, comparison, clinical notes, alerts with illustrative limits, and CSV export.

7. Scope and limits

Research prototype on cardboard insoles with a one-point calibration; the AI score is a screening indication trained on a different insole and is not a diagnosis. Not a medical device.